META-ANALYSIS

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Noninvasive Brain Stimulation in Attention-Deficit/Hyperactivity Disorder: A Systematic Review and Meta-Analysis of Treatment Efficacy and Targeting Strategies

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Objective: Noninvasive brain stimulation (NIBS) has emerged as a promising novel treatment for attention-deficit/hyperactivity disorder (ADHD). However, the precise effects of NIBS in ADHD as well as optimal stimulation protocols remain elusive. This systematic review and meta-analysis aimed to provide a rigorous evaluation and robust quantified evidence on the therapeutic efficacy of NIBS in ADHD.

Method: A systematic search of PubMed, Web of Science, and Embase identified 59 studies (repetitive transcranial magnetic stimulation [rTMS]: 11; transcranial direct current stimulation [tDCS]: 35; other NIBS: 13). Of these, 33 randomized controlled trial studies (7 for rTMS, 26 for tDCS; 1,049 participants with ADHD) were included in meta-analyses of therapeutic efficacy. Subgroup analyses were performed to investigate the impact of stimulation targets on therapeutic outcomes, and meta-regression analyses were included to evaluate the dose-response relationship.

Results: NIBS has favorable tolerability, but evidence for its therapeutic efficacy in ADHD remains inconclusive. Meta-analyses indicated that rTMS was effective in alleviating inattention symptoms (Hedges' $g = -1.44$, $p = .002$), reducing hyperactivity/impulsivity symptoms (Hedges' $g = -0.95$, $p = .005$), and improving attention-related functions (Hedges' $g = 0.53$, $p = .023$), and tDCS showed therapeutic benefits in reducing hyperactivity/impulsivity symptoms (Hedges' $g = -0.28$, $p = .024$) and improving inhibitory control (Hedges' $g = 0.39$, $p = .028$). tDCS positioned over the left dorsolateral prefrontal cortex and right orbitofrontal cortex demonstrated significant effects in improving inhibitory control. No significant dose-response relationship was found for either tDCS or rTMS in ADHD.

Conclusion: This meta-analysis supports the therapeutic potential of NIBS in ADHD, revealing domain-specific benefits across different stimulation devices and sites. Although more data are needed, NIBS shows promise as an intervention for ADHD.

Plain language summary: This systematic review and meta-analysis investigated the role of noninvasive brain stimulation in the treatment of attention-deficit/hyperactivity disorder (ADHD). Drawing on data from 33 randomized controlled trial studies with 1,049 participants, the authors found that repetitive transcranial magnetic stimulation (rTMS) was associated with improved attention and reduced hyperactivity/impulsivity, while transcranial direct current stimulation (tDCS) helped enhance inhibitory control and reduce hyperactivity/impulsivity. The effectiveness was linked to which brain region was stimulated. These results suggest that noninvasive brain stimulation may be a valuable treatment option for individuals with ADHD, particularly for those who do not respond well to standard medications.

Study registration information: Efficacy of Noninvasive Brain Stimulation and the Impact of Targeting in Attention Deficit/Hyperactivity Disorder: An Updated Systematic Review and Meta-Analysis; https://osf.io/bmhax

Diversity & Inclusion Statement: We worked to ensure sex and gender balance in the recruitment of human participants. We worked to ensure race, ethnic, and/or other types of diversity in the recruitment of human participants. We worked to ensure that the study questionnaires were prepared in an inclusive way. Diverse cell lines and/or genomic datasets were not available. We actively worked to promote inclusion of historically underrepresented racial and/or ethnic groups in science in our author group. We actively worked to promote sex and gender balance in our author group. While citing references scientifically relevant for this work, we also actively worked to promote inclusion of historically underrepresented racial and/or ethnic groups in science in our reference list. While citing references scientifically relevant for this work, we also actively worked to promote sex and gender balance in our reference list. The author list of this paper includes contributors from the location and/or community where the research was conducted who participated in the data collection, design, analysis, and/or interpretation of the work.

Key words: attention-deficit/hyperactivity disorder; noninvasive brain stimulation; transcranial magnetic stimulation; transcranial direct current stimulation; stimulation site

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Attention-deficit/hyperactivity disorder (ADHD) is characterized by age-inappropriate symptoms of inattention, hyperactivity, and impulsivity.[1] Pharmacotherapy (eg, psychostimulants) is currently the first-line treatment for ADHD,[2] but it may be associated with potential risks of misuse and addiction.[3] Further, individuals with ADHD may develop therapeutic tolerance following prolonged pharmacological treatment, rendering conventional medications less effective.[4] ADHD medications also have adverse effects that primarily stem from their nonspecific actions in the brain. For example, psychostimulants broadly modulate dopamine and norepinephrine neurotransmission across multiple neural circuits, rather than selectively affecting pathological pathways involved in specific ADHD symptom domains.[5]

Noninvasive brain stimulation (NIBS) includes a group of techniques that modulate brain activity externally and target specific brain regions that induce neuroplasticity.[6,7] NIBS has emerged as a promising therapeutic intervention in psychiatry, initially demonstrating efficacy in treating major depressive disorder through targeted stimulation of the left prefrontal cortex to encourage cortical inhibition necessary for adaptive mood regulation.[8] Building on this success, application of NIBS expanded to obsessive-compulsive disorder, where it alleviated symptoms by targeting dysfunction in prefrontal-striatal circuits.[9] Given these encouraging outcomes in modulating prefrontal function and associated neural networks, NIBS has been introduced as a potential therapeutic approach for ADHD. Commonly used methods for ADHD include transcranial magnetic stimulation (TMS), transcranial electrical stimulation (eg, transcranial direct current stimulation [tDCS]), and external trigeminal nerve stimulation (eTNS).[10] Although preliminary evidence suggests that NIBS may improve clinical symptoms and cognitive deficits in ADHD by targeted modulation of dysfunctional neural circuits,[11,12] empirical testing has yielded inconsistent findings,[13,14] necessitating further systematic investigation.

Previous reviews on the application of NIBS in ADHD have also yielded inconsistent conclusions.[15-20] The inconsistency may arise from the heterogeneity of participant characteristics, outcome measures, and experimental designs, which restricts the generalizability and comparability of findings. Moreover, prior research insufficiently explored how stimulus targets and parameters (eg, treatment duration) influence the efficacy of NIBS, limiting its utility in guiding clinical applications. With a notable increase in studies investigating NIBS in ADHD, an up-to-date meta-analysis incorporating a broad range of NIBS techniques and comprehensive measures of both clinical and cognitive outcomes is urgently needed.

Thus, our study aimed to synthesize the evidence from various NIBS techniques used in the treatment of ADHD and to provide a critical appraisal of their clinical and cognitive effects. Further, we conducted meta-analyses of tDCS and repetitive TMS (rTMS) studies in ADHD to quantitatively synthesize and compare cognitive and clinical outcomes. To elucidate the importance of intervention-specific factors, subgroup analyses stratified by anatomical stimulation site and meta-regression analyses on dose-response relationships were performed. Thus, this systematic review and meta-analysis will provide a rigorous evaluation and robust quantified evidence on the efficacy of NIBS in ADHD.

METHOD

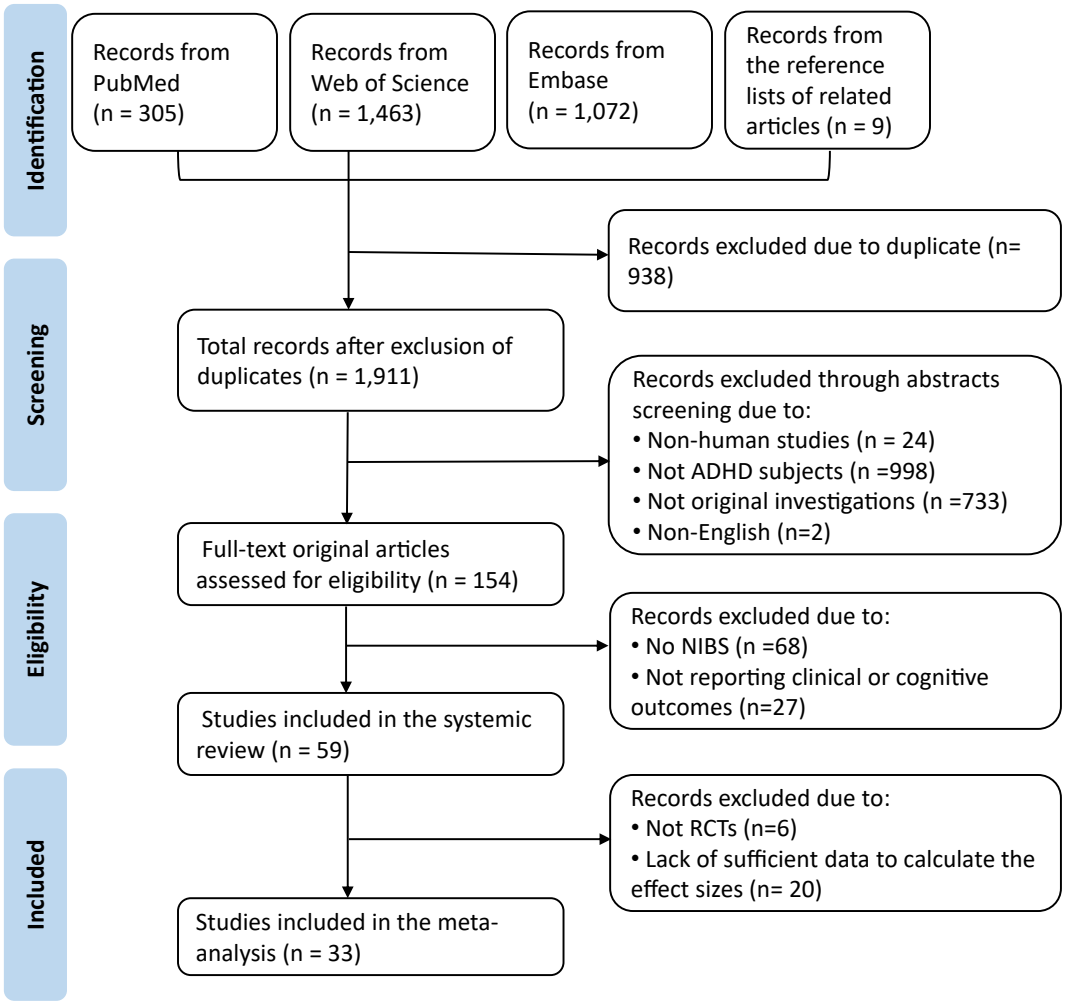
Literature Search and Eligibility Criteria

The study protocol was preregistered on the Open Science Framework (https://osf.io/bmhax). We conducted a comprehensive literature search in PubMed, Web of Science, and Embase for NIBS studies up to July 5, 2024 (detailed search strategy is provided in Supplement 1, available online). In addition, manual searches were performed of the reference lists of previous systematic reviews and meta-analyses. This systematic review and meta-analysis was conducted according to the Preferred Reporting Items for Systematic Evaluation and Meta-Analysis (PRISMA) criteria.[21] Studies comprising participants with ADHD confirmed by meeting either a clinical criterion of ADHD defined by *DSM*/*ICD* or cutoff criteria on validated ADHD rating scales and studies adopting any NIBS techniques were included. Studies were excluded if they did not examine clinical or cognitive outcomes of NIBS, were not original research, were written in non-English languages, or did not provide sufficient details on methods or results. Studies other than randomized controlled trials (RCTs) (for meta-analysis only) were also excluded. For studies with multiple publications from overlapping samples, the study with the largest sample was included in the meta-analysis.

Study Selection and Data Extraction

The search identified 59 studies (rTMS: 11; tDCS: 35; other NIBS: 13) in systematic reviews and 33 studies (7 for rTMS, 26 for tDCS) in meta-analyses of therapeutic efficacy. The study selection process is shown in Figure 1; details on studies excluded after full-text review are provided in Table S1, available online. We extracted comprehensive data from included studies, as follows:

FIGURE 1 Flowchart of Study Selection



Note: *ADHD = attention-deficit/hyperactivity disorder; NIBS = noninvasive brain stimulation; RCTs = randomized controlled trials. Please note color figures are available online.*

1. Study characteristics (authors, publication year, research country, sample size, study design)
2. Participant characteristics (age, sex, diagnostic criteria, ADHD subtype, comorbidities, medication status)
3. Intervention parameters (stimulation type, sham protocol, target location, targeting method, frequency/current, treatment duration, washout period for crossover designs)
4. Outcome measures (clinical symptom ratings, cognitive performance measures, neurophysiological indicators, and tolerability/adverse effects)

Two authors (Q.C. and Y.L.) separately carried out the search, performed the study selection and data extraction, and discussed discrepancies; a third author (N.P.) cross-checked the results and was consulted if consensus could not be reached.

Meta-analysis

Given the limited number of included studies (meta-analysis was performed when ≥3 studies were identified),[22] we conducted meta-analyses only on the clinical and cognitive effects of rTMS and tDCS in ADHD (details about calculation of Hedges' *g* are provided in Supplement 2, available online). To reduce heterogeneity, we clustered clinical symptoms and cognitive function into different domains and conducted independent meta-analyses for each domain. Specifically, clinical symptoms were divided into inattention and hyperactivity/impulsivity domains. Cognitive function consisted of 4 domains: attention, inhibitory control, processing speed, and working memory (categorization of cognitive outcomes is provided in Table S2, available online).

We conducted a random-effects meta-analysis to estimate the effect of NIBS by calculating Hedges' g values as the standardized mean difference in clinical and cognitive measures of ADHD. When multiple indicators were assessed using identical scales or tasks, the indicator that most accurately represented the primary construct was selected as the principal outcome variable based on the research objectives. For outcomes measured by multiple scales, means and standard deviations were pooled to calculate the overall effect size. If the same cohort underwent multiple experiments with different stimulus conditions (stimulation targets, current intensity), all conditions were included in the analysis. To avoid inflating the sample size, the total sample size of the cohort was proportionally divided across the different conditions included in the meta-analysis. The heterogeneity of effects was assessed using the I^2 statistic, which was categorized into low ($\leq$50%), moderate (50%-75%), and high ($\geq$75%) levels.[23]

To systematically identify potential sources of heterogeneity among the included studies, we conducted a narrative review highlighting the key variations across studies. In the subgroup analysis stratified by stimulation targets, the stimulation targets were generally assigned to 6 groups: anodal left dorsolateral prefrontal cortex (DLPFC), anodal right inferior frontal cortex (IFC), anodal right orbitofrontal cortex (OFC), cathodal right OFC, cathodal right DLPFC, and cathodal left DLPFC. In the subgroup analysis stratified by age group, participants were classified as children (<18 years old) or adults. In the subgroup analysis stratified by trial structure, studies were divided into parallel and crossover trials. The subgroup analyses were conducted only when at least 3 independent studies were available.[22] We also performed a meta-regression analysis to evaluate the dose-response relationship. We defined rTMS dose as number of total stimulus × duration of each stimulus, whereas tDCS dose was defined as session × duration of each session. Given the potential for both linear and nonlinear associations between dose and efficacy,[24] we used both linear and quadratic models and compared their performance with leave-one-out evaluation (Supplement 3, available online).

Jackknife sensitivity analyses (ie, repeating the same analysis each time and excluding different studies) were performed to evaluate the reproducibility of the findings. When 2 or more high-risk studies were included, we conducted a sensitivity analysis that excluded all high-risk studies. We used visual inspection of funnel plots and Egger test to assess publication bias. All analyses were 2-tailed and performed at α level .05 using Stata version 18.0 (StataCorp LLC, College Station, Texas).

Risk-of-Bias Assessment

We used the revised Cochrane risk-of-bias tool (RoB 2) to assess the quality of RCTs included in the meta-analyses.[25] The tool has 5 different domains: randomization process, deviations from intended interventions, missing outcome data, measurement of outcome, selection of reported result.[26] The overall risk-of-bias assessment was generated after individual evaluation of all 5 quality domains. Studies with low risk of bias for all domains were judged as "low risk." Studies with at least some concerns of bias in any area, but without high risk of bias for any domain, were judged as "some concerns." Studies with at least 1 high risk of bias in any area or some concerns for multiple domains were judged as "high risk."

RESULTS

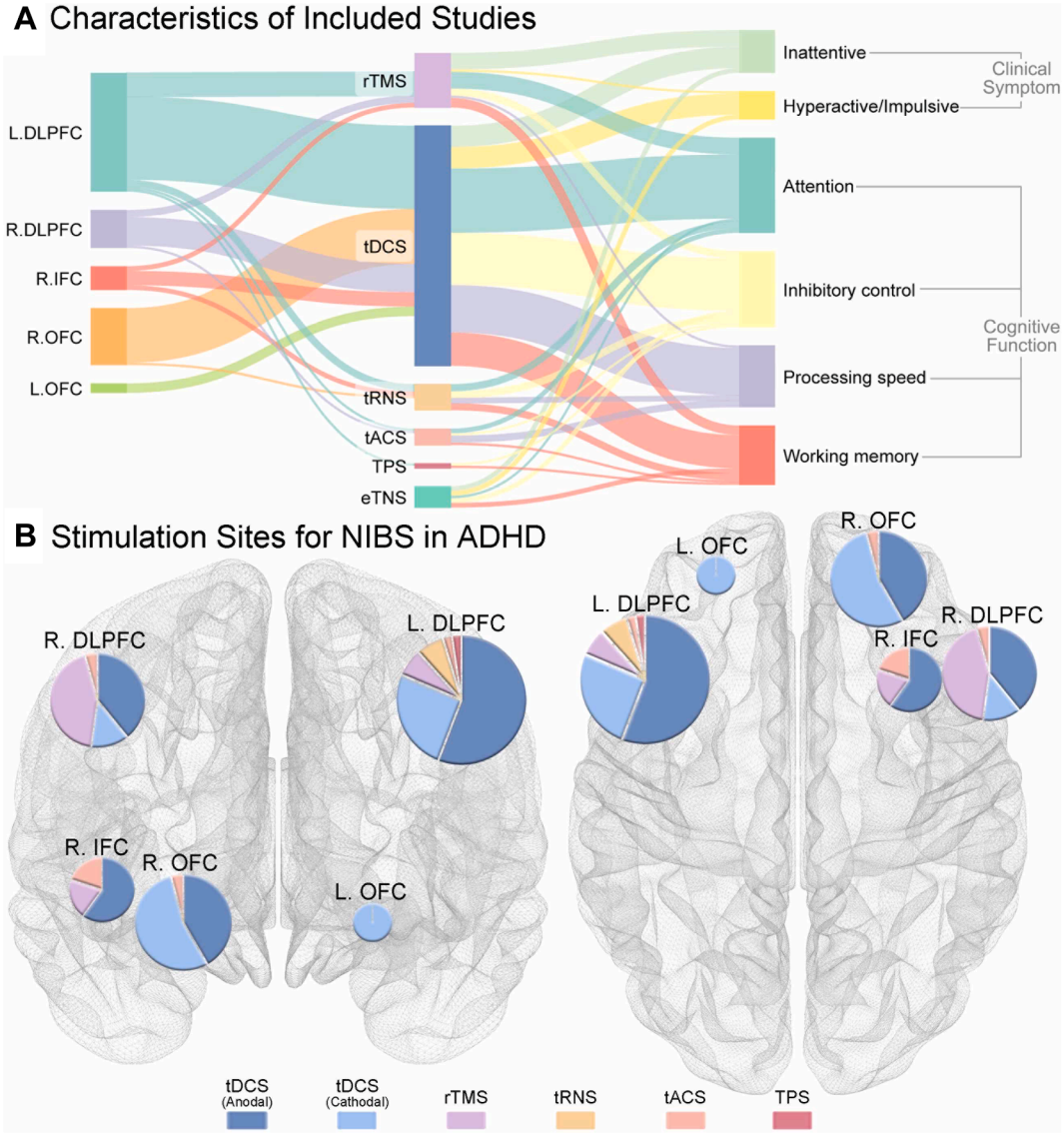
Sample Characteristics

This systematic review included 59 studies comprising 1,657 participants with ADHD (1,164 male participants, mean [SD] age = 17.08 [11.43] years). A total of 6 NIBS techniques were involved, including rTMS (11 studies), tDCS (35 studies), eTNS (2 studies), transcranial random noise stimulation (tRNS) (5 studies), transcranial alternating current stimulation (tACS) (5 studies), and transcranial pulse stimulation (TPS) (1 study) (Figure 2). Study and participant characteristics, intervention parameters, and outcome measures of individual studies are detailed in Table 1 and Table S3, available online.[11-14,27-80] The meta-analyses quantitatively evaluated the clinical and cognitive outcomes of rTMS and tDCS in ADHD, incorporating 7 RCTs of rTMS and 26 RCTs of tDCS, with 1,049 participants with ADHD (727 male participants, mean [SD] age = 19.91 [12.44] years).

Results of Systematic Review

rTMS Studies. Six studies indicated that rTMS could significantly improve core symptoms of ADHD such as inattention and hyperactivity/impulsivity.[27,29-33] In addition, 3 studies found that when rTMS is combined with pharmacotherapy or behavioral training, it can more effectively improve attention, hyperactivity/impulsivity symptoms, and sleep disorders.[11,34,35] These studies also observed significant enhancements in executive functions, such as arithmetic performance, coding ability, and emotional decision making.[11,34,35] Functional magnetic resonance imaging studies further revealed that rTMS targeting the right prefrontal cortex during working memory tasks enhanced brain network activity related to attention, including activation in the right DLPFC, right parietal cortex, and right inferior frontal gyrus, and

FIGURE 2 Overview of Noninvasive Brain Stimulation (NIBS) Applications in Attention-Deficit/Hyperactivity Disorder (ADHD) in Terms of Sites and Cognitive Constructs



Note: (A) A total of 59 NIBS studies were included in our systematic review and categorized by technology type: repetitive transcranial magnetic stimulation (rTMS), transcranial direct current stimulation (tDCS), external trigeminal nerve stimulation (eTNS), transcranial random noise stimulation (tRNS), transcranial alternating current stimulation (tACS), and transcranial pulse stimulation (TPS). The panel also recorded contributions to both clinical and cognitive assessments. (B) Brain regions are shown from both anterior and superior perspectives. The size of each pie chart represents the total number of studies targeting that specific brain region. Segment sizes within each pie chart indicate the proportion of different NIBS techniques applied to that region across the literature: dorsolateral prefrontal cortex (DLPFC) (left [L.]; F3 in electrocardiogram 10-20 system; right [R.]: F4); inferior frontal cortex (IFC) (R.: F8); orbitofrontal cortex (OFC) (L.: Fp1, R.: Fp2). Please note color figures are available online.

increased activation of the right DLPFC was significantly positively correlated with improvements in inattention and memory symptoms.[33] However, in 2 studies, although there was some improvement in overall clinical symptoms, the difference between the rTMS and sham groups was not statistically significant, and no improvement in attention function was observed (Table 1).[13,28] Among 8 studies that assessed the safety and tolerability of rTMS, 7 showed that rTMS was safe and well tolerated (Table S4, available online).[11,13,27-29,31,35] Common side effects included mild headaches, transient local discomfort, or mild pain in the treated area. Another study reported 1 case of seizures after rTMS.[32] Overall, rTMS has shown therapeutic efficacy for clinical symptoms across multiple studies, but evidence regarding its impact on cognitive function is limited. Individuals generally showed good tolerance toward rTMS,

TABLE 1 Clinical and Cognitive Effects in Noninvasive Brain Stimulation (NIBS) Studies

Study	Country	Sample size	Age, y		Sex, male		Stimulation protocol					Outcome measures		
			Mean	(SD)	n	(%)	Design	Stimulation target	Frequency/ current	Sessions	Session duration	Clinical measures	Cognitive measures	Other measures
rTMS														
Bloch *et al.*, 2010[27]	Israel	13	41.50	(13.57)	7	(53.85)	RCT, double-blind, crossover	Right DLPFC	20 Hz	1	1,680 pulses (2 s on, 30 s off)	PANAS (**total score**, mood, anxiety, hyperactivity); VAS (mood)	PANAS (**attention**); VAS (**attention**); CANTAB	—
Weaver *et al.*, 2012[28]	USA	9	18.11	(1.79)	6	(66.67)	RCT, single-blind, crossover	Right DLPFC	10 Hz	10	2,000 pulses (4 s on, 26 s off)	CGI-I; ADHD-IV	WAIS/WISC-IV; Connors CPT; DKEFS; Buschke Selective Reminding Test; Symbol Digit Coding test; finger oscillation tasks	—
Gomez *et al.*, 2014[29]	Cuba	10	9.50	(1.44)	10	(100.00)	Open-label, single-arm trial	Left DLPFC	1 Hz	5	1,500 pulses (on, off N/R)	ADHD symptoms checklist (**total score, inattention, hyperactivity/ impulsivity**)	—	—
Bayoumy *et al.*, 2014[30]	Egypt	41	9.39	(2.34)	25	(60.98)	Open-label, single-arm trial	Right DLPFC	1 Hz	5	1,200 pulses (on, off N/R)	**CPRSR-L**	—	EEG: **beta activity ISP latency**
Paz *et al.*, 2018[13]	Israel	rTMS: 9 Sham: 13	32.11 30.85	(6.47) (6.82)	14	(63.64)	RCT, double-blind, parallel	Both PFC	18 Hz	20	1,980 pulses (2 s on, 20 s off)	CAARS	TOVA (**total score**)	—
Cao *et al.*, 2018[11]	China	rTMS: 20 ATX: 19 rTMS + ATX: 21	8.36 9.22 8.50	(2.46) (2.39) (2.20)	49	(81.67)	RCT, double-blind, parallel	Right DLPFC	10 Hz	30	2,000 pulses (4 s on, 26 s off)	SNAP-IV (**attention deficit**, **hyperactivity impulse**, oppositional defiance)	CPT (**total score**); WISC (**digit span**, arithmetic, coding); **IGT**	—
Cao *et al.*, 2019[31]	China	rTMS: 18 Sham: 16 ATX: 21	8.83	(2.53)	34	(61.82)	RCT, double-blind, parallel	Right DLPFC	10 Hz	30	2,400 pulses (4 s on, 26 s off)	SNAP-IV (**attention deficit**, **hyperactivity impulse**, **oppositional defiance**)	—	**miRNA-let-7d**; miRNA-107
Alyagon *et al.*, 2020[32]	Israel	rTMS: 15 AC: 14 Sham: 14	26.62 26.13 27.64	(0.66) (0.59) (1.58)	34	(79.07)	RCT, double-semiblind, parallel	Right VLPFC and DLPFC	18 Hz	15	1,440 pulses (2 s on, 20 s off)	CAARS (**total score**, **hyperactivity/ impulsivity**); BAARS-IV (**hyperactivity/ impulsivity**); BDI	Stroop task (improvement); SST (SSRT); BRIEF-A	—
Bleich-Cohen *et al.*, 2021[33]	Israel	Right PFC: 24 Left PFC: 22 Sham: 16	35.60 35.10 34.70	(8.70) (10.10) (9.20)	40	(64.52)	RCT, double-blind, parallel	Right VLPFC and DLPFC	18 Hz	15	1,440 pulses (2 s on, 20 s off)	CAARS (total score, **inattention/ memory**); CGI; AAQoL; BDI	Mindstreams; BRIEF	fMRI: **right DLPFC**, **IPS**, **IFC** FC: **mPFC**, **PCC**
								Left VLPFC and DLPFC	18 Hz	15	1,440 pulses (2 s on, 20 s off)	CAARS; CGI; AAQoL; BDI	Mindstreams; BRIEF	fMRI: left DLPFC, IPS, IFC FC: mPFC, PCC

(*continued*)

TABLE 1 Continued

rTMS													
Nagy *et al.*, 2022[34]	Egypt	rTMS + ATX: 30 Sham + ATX: 30	8.47 8.70	(1.70) (1.76)	46	(76.67)	RCT, double-blind, parallel	Right DLPFC	10 Hz	15	2,000 pulses (4 s on, 26 s off)	CPRSR-L (**total score**, **inattention**, hyperactivity/impulsivity); **CGAS**; **CGI**	—	—
Yilin *et al.*, 2024[35]	China	rTMS + PBMT: 16 PBMT: 19	5.16 5.10	(1.90) (2.20)	22	(62.86)	Open-label, nonrandomized, parallel	Right DLPFC	1 Hz	12	1,000 pulses (10 s on, 10 s off)	**CSHQ**	—	—
tDCS														
Prehn-Kristensen *et al.*, 2014[36]	Germany	12	12.10	(1.40)	12	(100.00)	RCT, double-blind, crossover	Both DLPFC/both mastoid	0.25 mA	1	25 min	Macro sleep parameters; mood ratings	Declarative memory (**accuracy**); alertness; digit span task (correct)	—
Munz *et al.*, 2015[37]	Germany	14	12.30	(1.40)	10	(71.43)	RCT, double-blind, crossover	Both DLPFC/both mastoid	0.25 mA	1	25 min	Macro sleep parameters	Go/No-Go task (**RT**, **RT-V**, commission errors); intrinsic alertness (RT, RT-V); motor memory task (RT, accuracy)	**SO power**
Cosmo *et al.*, 2015[38]	Brazil	tDCS: 30 Sham: 30	31.83 32.67	(11.55) (10.37)	35	(58.33)	RCT, double-blind, parallel	Left DLPFC/right DLPFC	1 mA	1	20 min	—	Go/No-Go task (correct, commission errors, omission errors)	—
Breitling *et al.*, 2016[39]	Germany	21	14.33	(1.15)	21	(100.00)	RCT, single-blind, crossover	Right IFC/left mastoid	1 mA	1	20 min	—	Flanker task (incongruent trials: **commission errors**, **RT-V**, RT, omission errors)	—
Bandeira *et al.*, 2016[40]	Brazil	9	11.10	(2.80)	8	(88.89)	Single-arm trial, open-label	Left DLPFC/right OFC	2 mA	5	28 min	**PGI-I**	TAVIS-3 test of visual attention (**omission errors**); NEPSY-II: inhibitory control (**switch errors**); digit span task; Corsi cubes	—
Soff *et al.*, 2017[41]	Germany	15	14.20	(1.20)	12	(80.00)	RCT, double-blind, crossover	Left DLPFC/vertex	1 mA	5	20 min	FBB-ADHD (**inattention**, hyperactivity, **impulsivity**)	QbTest (**inattention**, **hyperactivity**, impulsivity)	—
Sotnikova *et al.*, 2017[42]	Germany	13	14.33	(1.32)	10	(76.92)	RCT, double-blind, crossover	Left DLPFC/vertex	1 mA	1	20 min	—	QbTest (**RT**, **RT-V**, *omission errors*, *accuracy*)	fMRI: **left DLPFC**, **PMC**, **S1**, **right PC** FCD: **left DLPFC**
Cachoeira *et al.*, 2017[43]	Brazil	tDCS: 9 Sham: 8	31.00 33.75	(6.17) (3.65)	12	(70.59)	RCT, double-blind, parallel	Right DLPFC/left DLPFC	2 mA	5	20 min	ASRS (**total score**, **inattention**, hyperactivity/impulsivity); **SDS**	—	—

(*continued*)

TABLE 1 Continued

tDCS														
Allenby *et al.*, 2018[44]	USA	37	31.70	(13.57)	26	(70.27)	RCT, double-blind, crossover	Left DLPFC/right OFC	2 mA	3	20 min	—	Conners CPT (**commission errors**, omission errors, RT); SST; N-back task	—
Jacoby and Lavidor, 2018[45]	Israel	20	22.75	(2.80)	11	(55.00)	RCT, single-blind, crossover	Both DLPFC/ cerebellum	1.8 mA	1	20 min	PANAS	MOXO CPT (**hyperactivity score**, attention RT, timing score, distractor RT)	—
Soltaninejad *et al.*, 2019[46]	Iran	20	16.10	(0.83)	14	(70.00)	RCT, single-blind, crossover	Left DLPFC/right OFC	1.5 mA	1	15 min	—	Go/No-Go task (**Go accuracy**, RT); Stroop test	—
								Right OFC/left DLPFC	1.5 mA	1	15 min	—	Go/No-Go task (**No-Go accuracy**, RT); Stroop test (accuracy, RT)	—
Kashani Khatib *et al.*, 2019[47]	Iran	tDCS: 12 Sham: 12	9.00	(1.15)	17	(70.83)	RCT, parallel	Left DLPFC/right OFC	1 mA	10	20 min	—	Go/No-Go task (**No-Go accuracy**); N-back task (**accuracy**)	—
Breitling *et al.*, 2020[48]	Germany	14	13.90	(1.90)	12	(85.71)	RCT, double-blind, crossover	Right IFC/left OFC	1 mA	1	20 min	—	N-back task (corrected hit rate)	ERPs: **N200**, **P300**
Nejati *et al.*, 2020 (experiment 1)[49] Nejati *et al.*, 2020 (experiment 1)[50]	Iran	15	10.00	(2.23)	15	(100.00)	RCT, double-blind, crossover	Left DLPFC/right DLPFC	1 mA	1	15 min	—	Go/No-Go task (accuracy, RT); N-back task (accuracy, **RT**, **boundary separation**, drift rate, starting point bias, nondecision time); Stroop task (**accuracy**, **RT**); WCST	—
Nejati *et al.*, 2020 (experiment 2)[49]	Iran	10	9.00	(1.80)	5	(50.00)	RCT, double-blind, crossover	Left DLPFC/right OFC	1 mA	1	15 min	—	Go/No-Go task; N-back task (**accuracy**, **RT**); Stroop task; WCST (**perseverative errors**, **completed categories**, **total errors**)	—
								Right OFC/left DLPFC	1 mA	1	15 min	—	Go/No-Go task (**No-Go accuracy**); N-back task; Stroop task; WCST (**perseverative errors**, **completed categories**, **total errors**)	—

(*continued*)

TABLE 1 Continued

tDCS														
Salehinejad *et al.*, 2020[51]	Germany	17	9.53	(1.50)	12	(70.59)	RCT, single-blind, crossover	Right PPC/left shoulder	1 mA	1	23 min	—	ANT (**orienting**, alerting, executive control); Stroop task (incongruent trials: *accuracy*, *RT*); SAT; Go/No-Go task (accuracy, RT)	—
Nejati *et al.*, 2021[52]	Iran	20	8.60	(1.56)	16	(80.00)	RCT, single-blind, crossover	Right OFC/left DLPFC	1 mA	1	15 min	—	CDDT (**K10**, **K20**, Kmean); BART (**SPB**, AV, UV)	—
								Left DLPFC/right OFC	1 mA	1	15 min	—	CDDT (K10, **K20**, Kmean); BART	—
Nejati *et al.*, 2020[50] Nejati *et al.*, 2021[53]	Iran	24	9.25	(1.53)	16	(66.67)	RCT, single-blind, crossover	Right DLPFC/left shoulder	1 mA	1	20 min	—	Flanker task; CTT (**CTI**); Go/No-Go task (**No-Go bias, nondecision time**)	—
Breitling-Ziegler *et al.*, 2021[54]	Germany	tDCS: 9	13.22 (2.39)		19	(86.36)	RCT, double-blind, parallel	Right IFC/square around the anode	0.5 mA	5	20 min	DISYPS-II (*total score*, *hyperactivity*, inattention, impulsivity)	N-back/No-Go task (**omission errors**, **RT-V**, commission errors, N-back hit rate, RT); flanker task (**RT-V**, RT, error rate, omission errors); spanboard task (**omission errors**, error rate)	ERPs: P3
		Sham: 13	13.54	(1.45)										
		tDCS: 11	12.27	(1.68)	18	(75.00)	RCT, double-blind, parallel	Right IFC/square around the anode	0.25 mA		20 min	DISYPS-II (total score, hyperactivity, inattention, impulsivity)	N-back/No-Go task (*commission error*); flanker task; spanboard task	—
		Sham: 13	13.54	(1.45)										
Dubreuil-Vall *et al.*, 2021[12]	Spain	37	37.35	(15.12)	18	(48.65)	RCT, double-blind, crossover	Left DLPFC/right OFC	2 mA	1	30 min	—	EFT (**incongruent RT**, accuracy); SST (**Go RT**, SSRT, accuracy)	ERPs: **P300** (**EFT incongruent trials**), N200, **P200** (**SST go trials**)
								Right DLPFC/left OFC	2 mA	1	30 min	—	EFT (incongruent RT, accuracy); SST (Go RT, SSRT, accuracy)	ERPs: **P300** (**EFT incongruent trials**), N200, **P200** (**SST go trials**)
Westwood *et al.*, 2021[14]	United Kingdom	tDCS + CT: 24	13.05	(1.98)	35	(70.00)	RCT, double-blind, parallel	Right IFC/left OFC	1 mA	15	20 min	ADHD-RS (*inattention*, *hyperactivity/impulsivity*); *Conners 3-P ADHD Index*	Go/No-Go task (accuracy); CPT (commission errors, omission errors); Simon task; TDT; MCT (commission errors, omission errors); WCST (perseverative errors); working memory task; verbal fluency task (accuracy)	—
		Sham + CT: 26	14.23	(2.06)										

(*continued*)

TABLE 1 Continued

tDCS														
Leffa *et al.*, 2022[55]	Brazil	tDCS: 25 Sham: 30	38.30	(9.60)	25	(45.45)	RCT, double-blind, parallel	Right DLPFC/left DLPFC	2 mA	28	30 min	CAARS (**inattention**, hyperactivity/impulsivity); BDI; BAI	BRIEF-A	—
Klomjai *et al.*, 2022[56]	Thailand	11	8.55	(0.65)	10	(90.90)	RCT, double-blind, crossover	Right OFC/left DLPFC	1.5 mA	5	15 min	—	Visual Go/No-Go task (**omission errors**, false alarm errors); auditory CPT (omission errors, false alarm errors)	qEEG: alpha (EEG with eyes open in right frontal area), beta, delta (visual Go/No-Go task in left frontal area), epsilon
Salehinejad *et al.*, 2022[57]	Germany	tDCS: 11 Sham: 11	8.86	(1.80)	11	(50.00)	RCT, single-blind, crossover	Both DLPFC/both shoulder	1.5 mA	1	15 min	—	Flanker task (**congruent and incongruent RT**); Go/No-Go task (accuracy, RT); N-back task (accuracy, RT); WCST (perseverative errors, completed categories, RT)	—
Schertz *et al.*, 2022[58]	Israel	tDCS + CT: 13 Sham + CT: 12	10.83	(1.79)	18	(72.00)	RCT, double-blind, parallel	Left DLPFC/vertex	1 mA	12	20 min	VADPRS	CBCL (**anxiety depression**, **social problems**); BRIEF (**surrounding disorder**); CANTAB	—
Barham *et al.*, 2022[59]	Turkey	tDCS: 11 Sham: 11	22.45 21.72	(3.00) (2.61)	7	(31.82)	RCT, double-blind, parallel	Right DLPFC/left DLPFC	2 mA	5	20 min	—	Digit span task (**backward**); ToL (**TMS**); Stroop task (interference time, false response); CPT (correct, omission errors); RMET; TMT	—
Nejati *et al.*, 2023 (experiment 1)[60]	Iran	18	8.83	(1.50)	14	(77.78)	RCT, single-blind, crossover	Left DLPFC/right OFC	2 mA	1	15 min	—	Verbal fluency test (**phonemic**, semantic)	—
								Right OFC/left DLPFC	2 mA	1	15 min	—	Verbal fluency test (phonemic, semantic)	—
Nejati *et al.*, 2023 (experiment 2)[60]	Iran	19	9.57	(1.38)	18	(94.74)	RCT, single-blind, crossover	Right DLPFC/left arm	2 mA	1	15 min	—	Verbal fluency test (phonemic, **semantic**)	—
								Right IFC/left arm	2 mA	1	15 min	—	Verbal fluency test (phonemic, **semantic**)	—

(*continued*)

TABLE 1 Continued

tDCS														
Nejati *et al.*, 2023[61]	Iran	23	10.09	(2.02)	22	(95.65)	RCT, single-blind, crossover	Left DLPFC/right OFC	1.5 mA	1	15 min	—	BART (**SP**, Phi, Eta); CDDT (S, K, Beta, R)	—
								Right OFC/left DLPFC	1.5 mA	1	15 min	—	BART (SP, Phi, *Eta*); CDDT (S, K, Beta, R)	—
								Left DLPFC/right shoulder	1.5 mA	1	15 min	—	BART (SP, **Phi**, Eta); CDDT (S, K, Beta, R)	—
								Right OFC/left shoulder	1.5 mA	1	15 min	—	BART (SP, Phi, Eta); CDDT (**S**, K, Beta, R)	—
Wang *et al.*, 2023[62]	China	HD-tDCS: 24 Sham: 23	11.29 11.74	(2.51) (2.59)	27	(57.45)	RCT, double-blind, parallel	Right OFC/square around the anode	1.0 mA	10	20 min	SNAP-IV (inattention, hyperactivity/ impulsivity); PSQI	IVA-CPT (**commission errors**, **false response number**, correct response number, omission, RT); TOH (**completion time**, completed steps); Stroop Color and Word Test	—
Weinberg *et al.*, 2023[63]	Israel	ADHD: 22; HC: 20	22.90	(2.71)	15	(35.71)	RCT, double-blind, parallel	Right IFC/left OFC	1 mA	1	10 min	—	SST (**SSRT**); control task (accuracy)	—
Basiri and Hadianfard, 2023[64]	Iran	tDCS:20 DBT:20 tDCS + DBT:20 Control:20	23.50	(2.90)	63	(78.75)	RCT, open-label, parallel	Left DLPFC/right DLPFC	1.5 mA	10	20 min	ASRS (**inattention**, hyperactivity); **DERS**	Stroop test (**scores**); CPT (**scores**)	—
								Left DLPFC/right DLPFC	1.5 mA	10	20 min	ASRS (**inattention**, **hyperactivity**); **DERS**	Stroop test (**scores**); CPT (**scores**)	—
D'Aiello *et al.*, 2023[65]	Iran	26	10.63	(1.41)	24	(92.31)	RCT, single-blind, crossover	Left DLPFC/right OFC	1 mA	1	20 min	—	SST (RT, accuracy); N-back task (N-back index)	—
Guimarães *et al.*, 2024[66]	Brazil	15	11.00	(3.10)	10	(66.67)	RCT, triple-blind, crossover	Left DLPFC/right OFC	2 mA	5	30 min	PGI-I	TAVIS-4 test of visual attention (commission errors, omission errors, RT); NEPSY-II: IR (errors, RT); digit span task; Corsi block-tapping task	—
Nejati and Estaji, 2024[67]	Iran	22	9.16	(1.63)	14	(63.64)	RCT, single-blind, crossover	Left DLPFC/right OFC	1.5 mA	1	20 min	—	Emotional Stroop task (**attention bias index**); DPT (attention bias index)	—
								Right OFC/left DLPFC	1.5 mA	1	20 min	—	Emotional Stroop task (**attention bias index**); DPT (attention bias index)	—

(*continued*)

TABLE 1 Continued

tDCS														
Estaji *et al.*, 2024[68]	Iran	24	9.16	(1.57)	18	(75.00)	RCT, single-blind, crossover	Left DLPFC/right OFC	2 mA	1	20 min	—	Emotional Go/No-Go task (**No-Go accuracy**, Go accuracy, Go RT); emotional 1-back task (**accuracy**, RT)	—
								Right OFC/left DLPFC	2 mA	1	20 min	—	Emotional Go/No-Go task (No-Go accuracy, Go accuracy, Go RT); emotional 1-back task (accuracy, RT)	—
Other NIBS studies														
McGough *et al.*, 2014[69]	USA	21	10.30	(2.10)	19	(90.48)	Single-arm trial, open-label	V1 branches of trigeminal nerve	120 Hz/2-4 mA	56	420-540 min	ADHD-RS (**inattention, hyperactivity/impulsivity**); **CGI**; **GCI-I**; **CDI**; CSHQ (**sleep anxiety, total bedtime problems, total sleep problems**); MASC	**BRIEF**; ANT (**incongruent RT**); SWMT	—
McGough *et al.*, 2019[70]	USA	eTNS: 32 Sham: 30	10.30 10.50	(1.40) (1.40)	40	(64.52)	RCT, double-blind, parallel	V1 branches of trigeminal nerve	120 Hz/2-4 mA	28	480 min	ADHD-RS (**total scores, inattention, hyperactivity/impulsivity**); **GCI-I**; MASC (**parent report**, child report); CGI; CSHQ; CDRS-R; ARI	BRIEF	qEEG: F4 (delta, theta, beta, gamma), Fz (gamma)
Berger *et al.*, 2021[71] Dakwar-Kawar *et al.*, 2022[72]	Israel	19	13.30	(1.90)	17	(89.47)	RCT, double-blind, crossover	tRNS: left DLPFC and right IFC tDCS: left DLPFC/right OFC	0.75 mA	5	20 min	ADHD-RS (**total scores**); CGI-S	MOXO CPT (**timing index**, **PS**); digit span task (**backward**); MOXO CPT (**PS**)	—
Dakwar-Kawar *et al.*, 2023[73]	Israel	tRNS + CT:11 Sham + CT:12	9.25 8.64	(1.42) (1.43)	20	(86.96)	RCT, double-blind, parallel	Left DLPFC and right IFC	0.75 mA	10	20 min	ADHD-RS (**total scores**); CGI-S; PSQI (**sleep onset latency**, **wake-up times**)	Digit span task (forward, backward); MOXO CPT (PS); BRIEF	Resting-state EEG: alpha, beta (**bandwidth**), theta
Nejati *et al.*, 2024[74]	Iran	18	9.89	(1.90)	12	(66.67)	RCT, single-blind, crossover	Left DLPFC and right OFC	1 mA	1	20 min	—	CTT (**total time**); Go/No-Go task (**No-Go accuracy**, Go accuracy, Go RT); 1-back task (**accuracy**, RT); WCST (cluster, perseveration error, **total time**); BART (**AV**, **SP**)	—

(*continued*)

TABLE 1 Continued

Other NIBS studies														
Dallmer-Zerbe *et al.*, 2020[75]	Germany	tACS: 9 Sham: 9	31.3	(9.89)	11	(61.11)	RCT, single-blind, parallel	C3, C4, CP3, CP4, P3, P4 and T7, T8, TP7, TP8, P7, P8	1 mA	1	20 min	—	VOT (**omission error**, RT-M, RT-V)	ERPs: **P300 amplitude**; ERSP
Farokhzadi *et al.*, 2020[76]	Iran	tACS: 31 Sham: 31	9.49	(2.30)	44	(70.97)	RCT, parallel	Left DLPFC and right DLPFC	10 Hz	24	15 min	—	IVA (**attention**, **response control**)	—
Kannen *et al.*, 2022[77]	Germany	19	27.95	(8.57)	9	(47.37)	RCT, crossover	C1/C2 and C5/C6	1 mA	1	20 min	—	VOT (commission errors, omission errors, RT, RT-V); d2 Test of Attention (commission errors, omission errors)	ERPs: P300, **N700**; EROs
Amouzadeh and Sheikh, 2022[78]	Iran	tACS: 15 Sham: 15 Control: 15	11.62 11.60 11.60	(0.35) (0.34) (0.34)	30	(66.67)	RCT, parallel	F3 and SO	10 Hz/1 mA	10	15 min	—	N-back task (**correct responses**, **RT**); Stroop test (**number of errors**, RT)	—
Kannen *et al.*, 2024[79]	Germany	15	32.53	(11.07)	11	(73.33)	RCT, crossover	Cz and Oz	1.5 mA	1	18 min	Subjective ADHD symptom evaluation	CPT (commission errors, omission errors, RT-V)	Eye tracking; actigraphy; EEG: alpha power
Cheung *et al.*, 2024[80]	China	TPS: 17 Sham: 15	12.8 13.3	(1.51) (1.34)	25	(78.12)	RCT, double-blind, parallel	Left DLPFC	4 Hz	6	30 min	**SNAP-IV**; ADHD-RS-IV; CGI (**total score**, **improvement**, severity, efficacy)	Stroop task (RT); digit span task (forward, backward)	—

Note: *Boldface indicates improvement; italic indicates impairment. AAQol = Adult ADHD Quality of Life Measure; AC = active control; ADHD = attention-deficit/hyperactivity disorder; ADHD-IV = ADHD Rating Scale IV; ADHD-RS = ADHD Rating Scale; ANT = Attention Network Task; ARI = Affective Reactivity Index; ASRS = Adult ADHD Self-Report Scale; ATX = atomoxetine; AV = adjusted value; BAARS-IV = Barkley Adult ADHD Rating Scale–IV; BAI = Beck Anxiety Inventory; Beta = inverse temperature; BDI = Beck Depression Inventory; BART = Balloon Analogue Risk-Taking Task; BRIEF = Behavior Rating Inventory of Executive Functioning; BRIEF-A = Behavior Rating Inventory of Executive Functioning–Adult Version; CAARS = Conners' Adult ADHD Rating Scale; CANTAB = Cambridge Neuropsychological Test Automated Battery; CBCL = Child Behavior Checklist; CDDT = chocolate delay discounting task; CDI = Children's Depression Inventory; CDRS-R = Children's Depression Rating Scale–Revised; CGAS = Children's Global Assessment Scale; CGI = Clinical Global Impression; CGI-I = Clinical Global Impression Improvement Scale; CGI-S = Clinical Global Impression Severity Scale; CPRSR-L = Conners' Parent Rating Scale–Revised Long version; CPT = continuous performance task; CSHQ = Children's Sleep Habits Questionnaire; CT = cognitive training; CTI= circle tracing index; CTT = circle tracing task; DBT = dialectical behavior therapy; DERS = Difficulties in Emotion Regulation Scale; DISYPS-II = Diagnostic System for Mental Disorders in Childhood and Adolescence–Second Edition; DKEFS = Delis-Kaplan Executive Function System; DLPFC = dorsolateral prefrontal cortex; DPT = dot probe task; EEG = electroencephalography; EFT = Eriksen Flanker Task; ERO = event-related oscillation; ERP = event-related potential; ERSP = event-related spectral perturbation; eTNS = external trigeminal nerve stimulation; FBB-ADHD = Fremdbeurteilungsbogen for Attention-Deficit/Hyperactivity Disorder; FC = functional connectivity; FCD = functional connectivity density; fMRI = functional magnetic resonance imaging; HC = healthy control; HD-tDCS = high-definition transcranial direct current stimulation; IFC = inferior frontal cortex; IGT = Iowa Gambling Task; IPS = intraparietal sulcus; IR = inhibiting response; IVA-CPT = Integrated Visual and Auditory Continuous Performance Test; K = hyperbolic discounting rate; MASC = Multidimensional Anxiety Scale for Children; MCT = Mackworth Clock Task; mPFC = medial prefrontal cortex; NEPSY-II = Developmental Neuropsychological Assessment–Second Edition; N/R = not reported; OFC = orbitofrontal cortex; PANAS = Positive and Negative Affect Schedule; PBMT = Parent Behavior Management Training; PC = precuneus; PCC = posterior cingulate cortex; PFC = prefrontal cortex; PGI-I = Patient Global Impression of Improvement; PMC = premotor cortex; PPC = posterior parietal cortex; PS = processing speed; PSQI = Pittsburgh Sleep Quality Index; qEEG = quantitative electroencephalography; R = exponential discounting rate; RCT = randomized controlled trial; RMET = Reading the Mind in the Eyes Test; RT = reaction time; RT-M = reaction time mean; rTMS = repetitive transcranial magnetic stimulation; RTV = reaction time variability; S = impatience; S1 = primary somatosensory cortex; SAT = Shifting Attention Test; SDS = Sheehan Disability Scale; SNAP-IV = Swanson, Nolan, and Pelham Rating Scale–Version IV; SO = supraorbital; SP = successful pumping; SPB = successfully pumped balloons; SSRT = stop-signal reaction time; SST = Stop Signal Task; SWMT = Spatial Working Memory Task; tACS = transcranial alternating current stimulation; tDCS = transcranial direct current stimulation; TDT = time discrimination task; TMS = total move score; TMT = Trail Making Test; TOH = Tower of Hanoi; ToL = Tower of London; TOVA = Test of Variables of Attention; TPS = transcranial pulse stimulation; tRNS = transcranial random noise stimulation; UV = unadjusted value; VADPRS = Vanderbilt ADHD Parent Rating Scale; VAS = visual analog scale; VLPFC = ventrolateral prefrontal cortex; VOT = visual oddball task; WAIS = Wechsler Adult Intelligence Scale; WCST = Wisconsin Card Sorting Task; WISC = Wechsler Intelligence Scale for Children; WISC-IV = Wechsler Intelligence Scale for Children, Fourth Edition.*

with reported adverse events predominantly classified as mild in severity.

tDCS Studies. Five studies reported that tDCS significantly alleviated clinical symptoms in individuals with ADHD, primarily inattention symptoms.[40,41,43,55,64] However, 4 studies did not find improvement of clinical symptoms with tDCS.[45,58,62,66] In addition, 2 studies indicated that tDCS targeting the right IFC had an adverse effect on overall clinical symptoms, inattention, and hyperactive/impulsive symptoms in individuals with ADHD.[14,54] In terms of cognitive function, evidence for tDCS efficacy in ADHD emerged across multiple domains: enhancement of attention processes was documented in 7 studies,[37,39,40,46,54,58,64] and improvement in inhibitory control was exhibited in 8 studies.[39,44-47,49,62] Further, 4 studies found significant improvements in processing speed,[12,37,42,49] and 4 found improvements in working memory performance.[47,49,54,59] Some studies also indicated that tDCS showed potential for improving reward processing, verbal fluency, and emotional regulation.[52,60,61,67,68] Seven studies did not observe significant improvements in cognitive functions following tDCS intervention, such as attention, inhibitory control, and working memory in ADHD.[14,38,48,53,55,65,66] Furthermore, 1 study reported that tDCS on the left DLPFC impaired attention function,[42] and another found that stimulation on the right posterior parietal cortex impaired inhibitory control (Table 1).[51]

Stimulation targets may influence the therapeutic effects of tDCS. Anodal tDCS targeting the left DLPFC and right OFC, as well as cathodal tDCS targeting the right OFC, has demonstrated improvement in certain cognitive functions across the majority of studies. In contrast, anodal tDCS targeting the right IFC and cathodal tDCS targeting the right and left DLPFC have shown inconsistent effects on cognitive functions. Findings from 5 studies, in which all experimental conditions were controlled except for the stimulation targets, revealed that different stimulation targets have varying impacts on cognitive function.[46,49,52,60,61] For instance, anodal tDCS on the left DLPFC effectively improved verbal fluency, whereas anodal tDCS on the right DLPFC and right IFC improved semantic fluency.[60] Anodal tDCS on the left DLPFC significantly enhanced attention and working memory, whereas anodal tDCS on the right OFC notably improved inhibitory control.[46,49]

Among 35 studies, 29 reported the safety and tolerability of tDCS. Among these, 26 studies showed that tDCS was safe and well tolerated (Table S4, available online).[12,37-42,44,45,48-55,57,59-62,65,66,68] Common mild adverse effects include mild skin redness, scalp burning, and tingling pain. However, 3 studies also reported that 5 of these participants experienced serious adverse events after receiving tDCS, including mood swings, decreased appetite, sleep disorders, and severe headaches.[14,43,58]

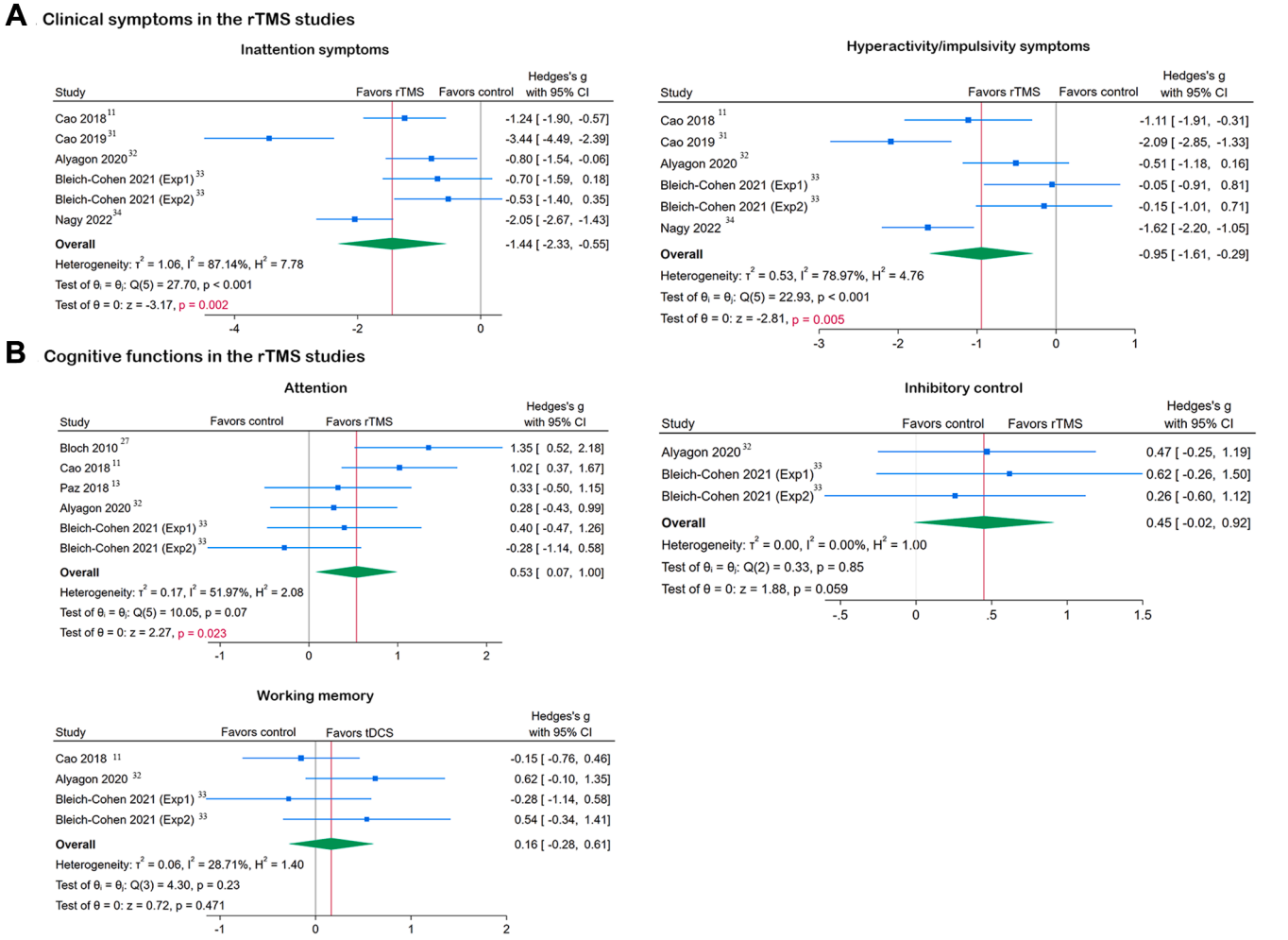
In summary, although tDCS has shown favorable safety and tolerability, definitive evidence for its therapeutic efficacy in improving clinical symptoms and cognitive function in individuals with ADHD remains inconclusive due to mixed results across studies. Moreover, cognitive outcomes appear to be differentially modulated by different stimulation sites.

Other NIBS Studies. eTNS (n =2) has shown significant efficacy in reducing inattention, hyperactivity, and impulsivity symptoms, as well as enhancing inhibitory control of individuals with ADHD.[69,70] Further, the clinical outcomes of eTNS were associated with increased electroencephalography activity in right frontal lobe and midfrontal line regions.[70] tRNS (n = 5) studies suggest potential therapeutic benefits in both reconciling clinical symptoms and enhancing executive functions, particularly inhibitory control and working memory, in individuals with ADHD and showed superior efficacy compared with tDCS.[71-74] tRNS also improves the processing speed of individuals with ADHD when faced with cognitive fatigue and environmental distractions.[72] Among the 5 included tACS studies, 3 confirmed its potential modulating effects on cognitive performance, such as attention, inhibitory control, and working memory. Importantly, observed cognitive enhancement exceeded the therapeutic efficacy of methylphenidate in both magnitude and duration.[75,76,78] However, the remaining 2 studies did not find that tACS can improve attention of individuals with ADHD.[77,79] Only 1 study has explored the effect of TPS, finding that TPS can significantly reduce the severity of symptoms in individuals with ADHD, with improvement effects persisting during the follow-up period of 1 to 3 months (Table 1).[80] These NIBS techniques were safe and well tolerated, and no serious adverse events were reported (Table S4, available online).

Results of Meta-analyses

Meta-analysis on Clinical and Cognitive Outcomes of rTMS. Meta-analyses showed significant improvement in inattention symptoms (Hedges' $g = -1.44$, 95% CI [−2.33, −0.55], $p = .002$; $I^2 = 87.14\%$, $p < .001$) (Figure 3A) and hyperactivity/impulsivity symptoms (Hedges' $g = -0.95$, 95% CI [−1.61, −0.29], $p = .005$; $I^2 = 78.97\%$, $p < .001$) (Figure 3A). Significant enhancement was observed in attention-related function (Hedges' $g = 0.53$, 95% CI [0.07, 1.00], $p = .023$; $I^2 = 51.97\%$, $p = .07$) (Figure 3B). In contrast, no significant improvement in

FIGURE 3 Therapeutic Efficacy of Repetitive Transcranial Magnetic Stimulation (rTMS)

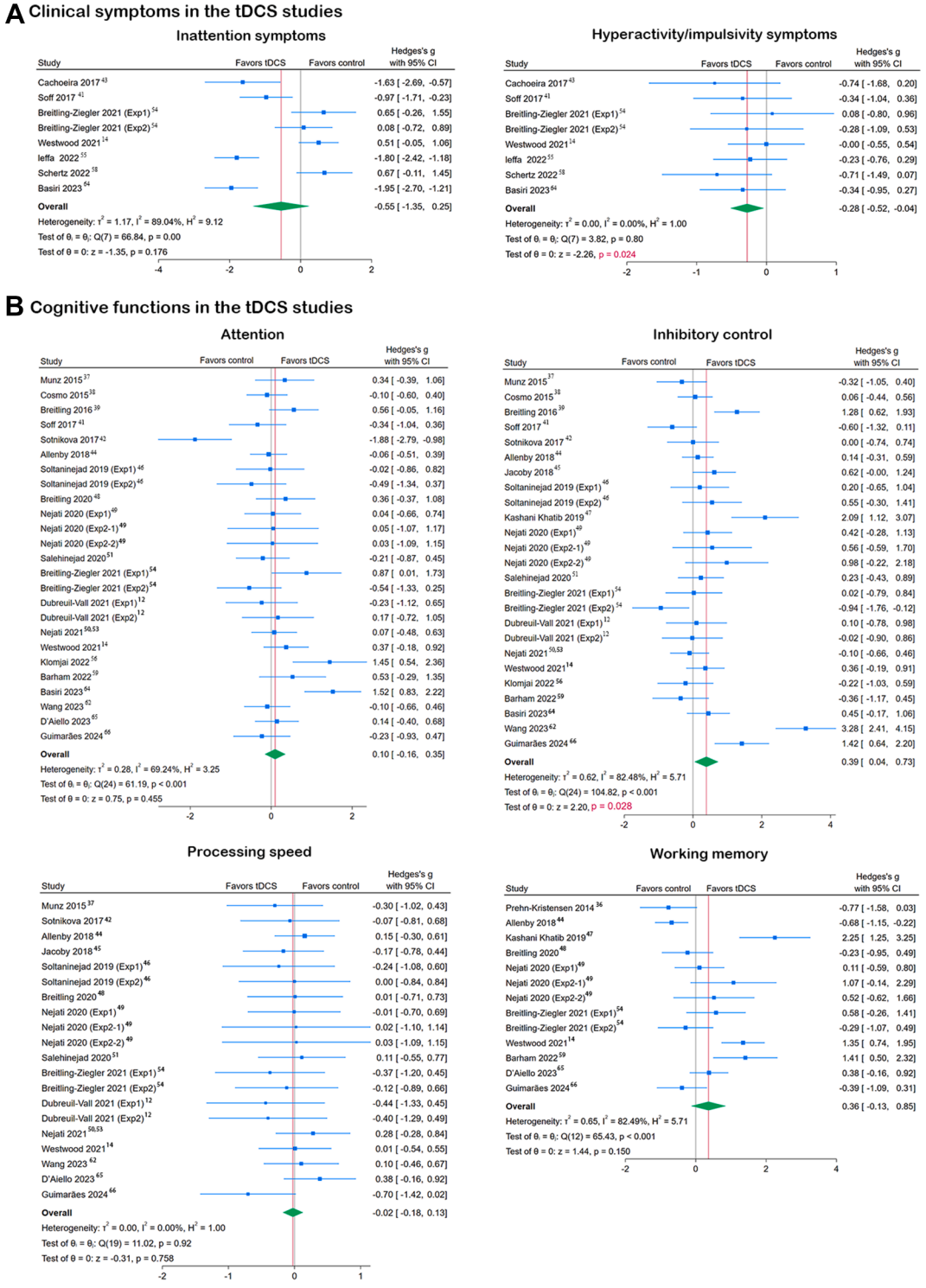


Note: *Forest plot of effect size for comparing the difference in improvement of clinical symptoms and cognitive functions between rTMS and control groups. (A) Clinical symptoms in the rTMS studies. (B) Cognitive functions in the rTMS studies. tDCS = transcranial direct current stimulation. Please note color figures are available online.*

inhibitory control (Hedges' $g = 0.45$, 95% CI [−0.02, 0.92], $p = .059$; $I^2 = 0.00\%$, $p = .85$) (Figure 3B) or working memory (Hedges' $g = 0.16$, 95% [CI −0.28, 0.61], $p = .471$; $I^2 = 28.71\%$, $p = .23$) (Figure 3B) was observed. Jackknife sensitivity analyses yielded consistent results, suggesting the robustness of our findings (Figure S1, available online). Assessment of publication bias through funnel plot analysis revealed no significant asymmetry (funnel plots in Figure S2, available online).

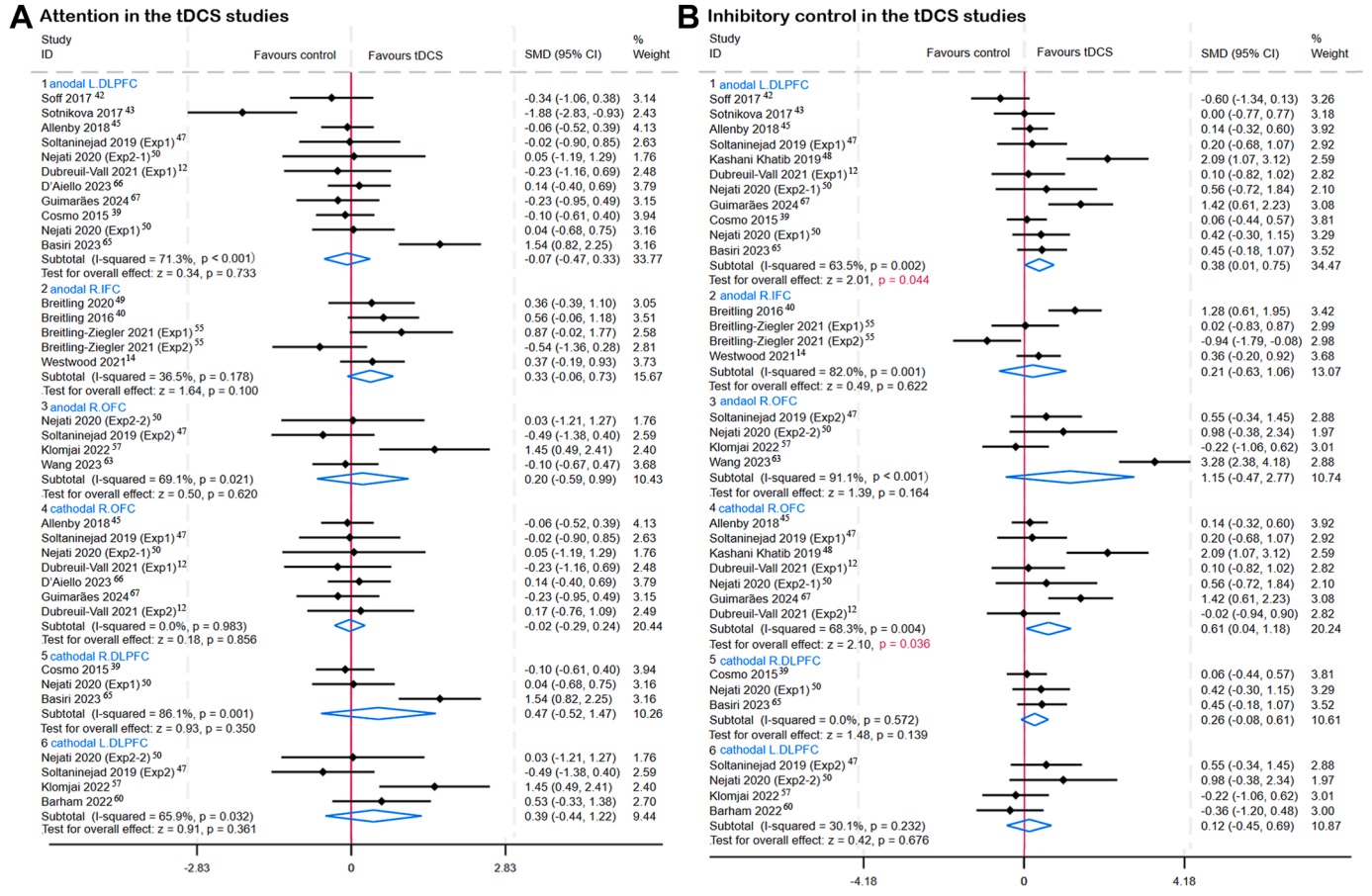
Meta-analysis on Clinical and Cognitive Outcomes of tDCS. Meta-analyses revealed no significant improvement in inattention symptoms (Hedges' $g = -0.55$, 95% CI [−1.35, 0.25], $p = .176$; $I^2 = 89.04\%$, $p < .001$) (Figure 4A). In contrast, significant improvement of hyperactive/impulsive symptoms (Hedges' $g = -0.28$, 95% CI [−0.52, −0.04], $p = .024$; $I^2 = 0.00\%$, $p = .80$) was demonstrated (Figure 4A). Significant enhancement was observed in inhibitory control (Hedges' g =0.39, 95% CI [0.04, 0.73], $p = 0.028$; $I^2 = 82.48\%$, $p < .001$) (Figure 4B). In contrast, there was no significant improvement in attention (Hedges' $g = 0.10$, 95% CI [−0.16, 0.35], $p = .455$; $I^2 = 69.24\%$, $p < .001$) (Figure 4B), processing speed (Hedges' $g = -0.02$, 95% CI [−0.18, 0.13], $p = .758$; $I^2 = 0.00\%$, $p = .92$) (Figure 4B), or working memory (Hedges' $g = 0.36$, 95% [CI −0.13, 0.85], $p = .150$; $I^2 = 82.49\%$, $p < .001$) (Figure 4B). Jackknife sensitivity analyses and sensitivity analyses that exclude high-risk studies yielded consistent results, suggesting the robustness of our findings (Figures S1 and S7, available online). Assessment of publication bias through funnel plot analysis revealed no significant asymmetry (funnel plots are provided in Figure S2, available online).

FIGURE 4 Therapeutic Efficacy of Transcranial Direct Current Stimulation (tDCS)



Note: *Forest plot of effect size for comparing the difference in improvement of clinical symptoms and cognitive functions between tDCS and control groups. (A) Clinical symptoms in the tDCS studies. (B) Cognitive function in the tDCS studies. Please note color figures are available online.*

FIGURE 5 Target-Specific Therapeutic Effects of Transcranial Direct Current Stimulation (tDCS)



Note: *Forest plot of effect sizes in subgroup analysis for cognitive functions in tDCS studies. (A) Attention in the tDCS studies. (B) Inhibitory control in the tDCS studies. DLPFC = dorsolateral prefrontal cortex; IFC = inferior frontal cortex; L. = left; OFC = orbitofrontal cortex; R. = right; SMD = standardized mean difference. Please note color figures are available online.*

Sources of Heterogeneity

Our included studies exhibited considerable heterogeneity, which mainly arose from differences in stimulation protocols, participant characteristics, and study designs. First, stimulation protocols may serve as a heterogeneity source, with diverse stimulation targets, treatment durations (ranging from a single session to 8 weeks) and intervention sessions (1-56 sessions). Second, variability in participant characteristics, particularly the imbalanced age distribution (67% children, 33% adults), contributes to the heterogeneity. Besides, inconsistency in diagnostic criteria across studies further compounds this variation (90% based on *DSM-IV*/*DSM-5* or the Kiddie Schedule for Affective Disorders and Schizophrenia for School-Age Children [K-SADS], 5% using the ADHD rating scale, and 5% not specified). ADHD presentations vary, as do comorbidity managing strategies (85% excluded psychiatric comorbidities, 12% included specific comorbidities, and 3% unreported). Medication status differs across studies: 63% required discontinuation of ADHD medications during treatment, 14% included only medication-naïve individuals, 5% allowed continued medication use, and 18% did not report medication status. Third, study design differences further heighten the heterogeneity, including variations in trial structure (40% parallel trials, 53% crossover trials, and 7% single-arm trials), intervention strategies (85% monotherapy treatment, 15% augmented treatment), washout periods in crossover trials (ranging from 1 day to 1 month), and outcome measures.

Results of Subgroup Analysis and Meta-regression

In the subgroup analysis of different stimulation sites, we examined the effects of tDCS on cognitive function, with a particular focus on attention and inhibitory control, as sufficient literature was available to support this analysis. tDCS with anodal stimulation over the left DLPFC and

cathodal stimulation over the right OFC showed significant enhancement in inhibition control in individuals with ADHD (Hedges' $g = 0.38$, 95% CI [0.01, 0.75], $p = .044$, and Hedges' $g = 0.61$, 95% CI [0.04, 1.18], $p = .036$, respectively) (Figure 5B), whereas other electrode montages failed to reach statistical significance ($p > .05$). Across all target regions, none of the stimulation protocols showed significant improvements in measures of attention (all $p > .05$) (Figure 5A).

In the subgroup analysis stratified by age groups, rTMS showed significant improvement in hyperactivity/impulsivity symptoms in children with ADHD (Hedges' $g = -1.62$, 95% CI [−2.17, −1.07], $p < .001$) (Figure S3B, available online), whereas no significant effect was found in adults. tDCS showed significant enhancement in inhibitory control in children with ADHD (Hedges' $g = .49$, 95% CI [0.02, 0.96], $p = .039$) (Figure S4B, available online), whereas no such enhancement was observed in adults. Conversely, tDCS was effective in alleviating inattention symptoms in adults with ADHD (Hedges' $g = -1.82$, 95% CI [−2.26, −1.39], $p < .001$) (Figure S4A, available online), but no significant effect was observed in children. Additionally, no significant differences were found between children and adults in the effects of tDCS on hyperactivity/impulsivity symptoms, attention-related function, or processing speed. In the subgroup analysis stratified by trial structure, tDCS showed a significant improvement in inhibitory control in individuals with ADHD in crossover trials (Hedges' $g = 0.28$, 95% CI [0.03, 0.54], $p = .029$) (Figure S5B, available online), but no significant effect was observed in parallel trials. In contrast, tDCS showed a significant improvement in working memory in parallel trials (Hedges' $g = 1.04$, 95% CI [0.20, 1.87], $p = .015$) (Figure S5D, available online), whereas no such effect was observed in crossover trials. No significant differences were found in the effects of study design on tDCS treatment outcomes for attention or processing speed.

The meta-regression analysis was performed to evaluate the relation between dose and efficacy in tDCS and rTMS studies on clinical and cognitive outcomes. Analysis of the relation between dose and efficacy using both linear and quadratic modeling yielded nonsignificant results in both rTMS (modulatory effects of dose and model comparisons are provided in Tables S5 and S6, available online) and tDCS studies (Tables S7 and S8, available online).

Risk of Bias

Among the eligible studies, 54.5% were assessed as having a low risk of bias, 36.4% had some concerns, and 9.1% were at high risk. Owing to lack of relevant information, 21.2% were considered to have some risk of bias in the randomization process, and 39.4% were considered to have some risk in selection bias when reporting results. Overall, the bias was low in the majority of domains (Figure S6, available online).

DISCUSSION

We provide a comprehensive and systematic review examining the therapeutic efficacy of NIBS techniques in ADHD. Meta-analyses demonstrated distinct therapeutic benefits for different NIBS modalities: rTMS significantly improved core clinical symptoms and enhanced attention-related cognitive functions, whereas tDCS effectively reduced hyperactivity/impulsivity symptoms and enhanced cognitive aspects of inhibitory control. Notably, tDCS with anodal stimulation over the left DLPFC and cathodal stimulation over the right OFC showed significant enhancement in inhibitory control, although treatment dose did not significantly influence cognitive outcomes. Compared with previous reviews and meta-analyses,[15-20] our study incorporated a larger sample size, more comprehensive outcome measures, and exploratory analyses of stimulation targets and dose-response relationships. These findings support the therapeutic potential of NIBS as a promising nonpharmacological intervention for ADHD and offer valuable insights for optimizing clinical application.

Our systematic review indicated that NIBS has the potential to improve clinical symptoms and certain cognitive functions in individuals with ADHD. However, the results of existing studies showed considerable heterogeneity, making it challenging to draw a definitive conclusion. This heterogeneity stems from 3 main factors: stimulation protocols (eg, stimulation targets may have a direct influence on treatment effects[14,41]), participant characteristics (eg, age may modulate different neuroplasticity responses), and study design differences (eg, potential learning effects in crossover designs may affect the accurate assessment of treatment effects).[45]

Despite the observed heterogeneity, our findings indicated that NIBS is well tolerated and safe. Additionally, the therapeutic effects of NIBS on ADHD can persist during the follow-up period. Two rTMS studies indicated that following 3 weeks of 15 treatment sessions, improvements in attention and hyperactivity/impulsivity can be maintained for up to 1 month, though with some attenuation over time.[31,34] Studies on tDCS suggested that short-term interventions (5-10 sessions) can sustain improvements in attention and inhibitory control for 2 to 6 weeks,[43,47,61] with 1 study suggesting that effects may last up to 4 months.[54] However, 1 study reported that the benefits of 5 tDCS sessions on inhibitory control did not persist at the

1-month follow-up.[56] Furthermore, other NIBS techniques, such as tACS and TPS, have also demonstrated varying degrees of long-term efficacy.[76,78,80] Based on the current evidence, NIBS may serve as an effective complement to ADHD management, particularly for individuals who may be refractory to or unsuitable candidates for pharmacological interventions.

Meta-analyses indicated that rTMS significantly improved core clinical symptoms and enhanced attention-related cognitive functions in individuals with ADHD, whereas tDCS can enhance inhibitory control functions and alleviate hyperactivity/impulsivity symptoms. rTMS generates a time-varying magnetic field in the targeted region to stimulate the cerebral cortex locally, modulating inhibitory and excitatory synaptic transmission and thereby regulating cortical excitability.[81] In ADHD treatment, the right DLPFC, a neural substrate crucial for attentional control, serves as the primary therapeutic target.[82] A study showed that rTMS intervention significantly enhances right DLPFC activation, with these neurophysiological changes exhibiting robust correlations with improvements in attention and memory functions.[33] This finding further confirms the critical role of the right DLPFC in attention regulation. Some studies have also reported that rTMS improved hyperactive/impulsive symptoms, especially when stimulation targets are extended to other regions of the prefrontal cortex, such as the ventrolateral prefrontal cortex and the left DLPFC.[29,32] In contrast, tDCS applies a constant, low-intensity direct current via electrodes, which influences voltage-gated ion channels and alters the resting membrane potential of neurons, modulating excitability thresholds.[83] tDCS typically employs a dual-electrode system comprising an anode and cathode, which not only modulates local neural activity at the stimulation site but also influences the intrinsic functional connectivity patterns within broader cortical networks.[42] In ADHD intervention, electrode placement primarily targets the prefrontal cortex, specifically engaging the prefrontal-striatal-thalamic circuit—a neural network fundamental to inhibitory control mechanisms.[84,85] This stimulation approach enhances both prefrontal cortical activity and its functional integration with connected neural substrates, thereby strengthening inhibitory control capabilities and subsequently ameliorating hyperactivity and impulsivity symptoms in patients with ADHD.

Further subgroup analysis stratified by stimulation targets revealed that tDCS applying anodal stimulation to the left DLPFC and the cathodal stimulation to the right OFC showed significant improvement in inhibitory control. This may be closely related to the roles of these 2 regions within the inhibitory control function. Inhibitory control, a core deficit in ADHD,[86] is primarily regulated by the prefrontal-striatal-thalamic circuit, with the DLPFC playing a crucial role.[87] Research has shown that individuals with ADHD exhibit significant structural and functional deficits in the DLPFC, and during inhibitory control tasks, the activation of the DLPFC is also significantly reduced.[88-91] At the same time, abnormalities in the OFC have been extensively observed in individuals with ADHD, including reduced gray matter volume and altered activation patterns.[92-94] These OFC dysfunctions in ADHD have been associated with impulsivity symptoms, which are closely linked to inhibitory control.[95] In addition, studies suggested that individuals with ADHD display abnormal functional connectivity between the DLPFC and related regions during inhibitory tasks,[96] and such network-level dysregulation may be a key contributor to deficits in inhibitory control. tDCS may enhance functional activity in the DLPFC and OFC while modulating the functional connectivity of relevant neural circuits, thereby enhancing inhibitory control function.

Subgroup analysis stratified by age groups revealed an age-dependent pattern of NIBS efficacy in the treatment of ADHD. rTMS showed a significant improvement in hyperactive/impulsive symptoms only in children's samples. Our findings suggest that clinical approaches to neurostimulation should be tailored according to age groups when targeting different symptoms. The efficacy of rTMS primarily in pediatric populations suggests that children's brains retain greater neuroplasticity in response to electromagnetic stimulation. In contrast, tDCS, with its dual-electrode configuration, likely engages more complex neural pathways, necessitating further research to determine its age-dependent effectiveness. Subgroup analysis stratified by trial structure revealed that study design may influence the effects of tDCS on cognitive function in individuals with ADHD due to heterogeneity in the washout periods across crossover trials, which may have introduced learning effects or carryover effects.[45] Therefore, future studies should prioritize crossover designs with an adequate washout period or consider using parallel-group designs to enhance study reliability.

Meta-regression analysis revealed that dose did not significantly impact efficacy in ADHD. The findings carry important clinical implications: rather than focusing on extending treatment dose, clinicians should prioritize other factors, such as the selection of stimulation targets corresponding to clinical profiles and stimulation intensity. Notably, the observed clinical and cognitive improvements even with smaller doses suggest that NIBS offers both enhanced treatment acceptability and clinical flexibility. However, these conclusions should be interpreted with caution. Our study considered only the session number and duration to investigate the dose-response relationship,

which are consistently reported across studies with superior generalizability. This approach, though methodologically sound for meta-analytic synthesis, provides only a partial characterization of dose dimensions. This limitation emphasizes the necessity for future studies specifically designed to investigate the dose-response relationship of NIBS.

Our study should be considered in light of some limitations. First, methodological heterogeneity across studies, including unstratified participant characteristics, diverse stimulation protocols, and varying study designs, limits the systematic evaluation of NIBS efficacy across studies conducted in ADHD. Second, the limited number of studies and sample sizes prevented comprehensive subgroup analyses of stimulation targets and made it difficult to establish a reliable dose-response relationship for different NIBS techniques, thus constraining the generalizability of our findings. Finally, the heterogeneity in domain-specific therapeutic outcomes across studies presents an important methodological consideration: these differences may be attributed either to the distinct mechanisms of various stimulation devices or to the differential selection of stimulation sites across devices. This confound limits our ability to determine the precise source of treatment specificity. Future research should aim to reconcile these sources of heterogeneity to enhance the specificity and robustness of the results.[97] Moreover, increased sample sizes and incorporating a broader range of NIBS techniques could facilitate personalized applications[98] and facilitate exploration of the comparative advantages of different NIBS technologies and optimal stimulation pipelines.

NIBS is a promising therapeutic approach to alleviating clinical symptoms and enhancing cognitive functions in individuals with ADHD. Specifically, rTMS demonstrates efficacy in improving core clinical symptoms and enhancing attention-related cognitive functions, whereas tDCS shows particular promise in reducing hyperactivity/impulsivity symptoms and strengthening inhibitory control. The therapeutic efficacy of NIBS interventions is notably influenced by stimulation protocols, with target region selection playing a crucial role. In particular, tDCS protocols incorporating anodal stimulation of the left DLPFC combined with cathodal stimulation of the right OFC have demonstrated significant improvements in inhibitory control among individuals with ADHD. Moving forward, future research should focus on optimizing NIBS protocols and investigating the comparative advantages of different NIBS technologies to establish their relative efficacy in addressing both clinical symptoms and cognitive deficits, ultimately advancing the practical therapeutic application of NIBS for ADHD management.

CRediT authorship contribution statement

Qiuxing Chen: Writing – original draft, Visualization, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Nanfang Pan:** Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization, Writing – original draft, Software. **Yajing Long:** Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Kun Qin:** Writing – review & editing. **Arshiya Sangchooli:** Writing – review & editing, Supervision, Conceptualization. **Ying Cao:** Methodology, Data curation. **Manpreet K. Singh:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Ying Chen:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Qiyong Gong:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

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